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THREAD ZERO

- **Problem Statement ID – PS26105**
- **Problem Statement Title - AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform**
- **Theme – Cybersecurity and Blockchain**
- **PS Category - Software**
- **Team ID – 150145**
- **Team Name – Thread Zero**



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MARGIN CALL

From static severity scores → continuous, financial cyber-risk decisions

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Margin Call translates complex cyber vulnerabilities into financial risk in Rupees/USD/ETC, mathematically selecting the highest-impact fixes that fit your budget while automating regulatory compliance.

THE PROBLEM

1. **Risk visibility goes stale**
   Manual audits miss real-time exposure.
2. **Risk isn't financial**
   CVSS says severity, not ₹ loss.
3. **Vague heatmaps**
   Leaders need clear, discussable numbers.
4. **Security budgets are guesswork**
   Boards lack a quantified ROI basis.

HOW OUR PLATFORM ADDRESSES IT (Uniquely)

FINANCIAL
QUANTIFICATION
CVEs → Expected Annual Loss (EAL)
+ 95% VaR in ₹

CONTINUOUS
AUTOMATION
Live scanner data + daily
EPSS / CISA KEV feeds

MATHEMATICAL
OPTIMIZATION
MILP / 0-1 Knapsack → fund
Highest-impact fixes within budget

Proposed Solution

- Margin Call turns complex cybersecurity warnings into simple financial decisions that business leaders can actually understand. Instead of handing executives confusing technical severity ratings, the platform calculates the potential monetary damage of a cyber attack in clear Indian Rupees.
- It runs thousands of breach scenarios to show exactly how much money is on the line, then automatically figures out the best way to spend a limited security budget to stop the biggest financial threats first.
- At the same time, an AI assistant creates simple compliance reports that prove the company follows official financial rules, while an intuitive dashboard lets managers slide a budget control to see what gets fixed and assign those tasks straight to their tech team.

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TECHNICAL APPROACH

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TECHNICAL FLOW

1 DATA INGESTION PIPELINE
Nessus / Qualys CVEs + EPSS + CISA KEV + Asset CMDB telemetry

↓

2 FAIR QUANTITATIVE MODELING
Poisson(λ) threat frequency + Binomial control strength + Beta-PERT loss magnitude + 10,000 Monte Carlo runs

↓

3 DECISION ENGINE
PuLP 0/1 Knapsack + multi-objective spend tuning + RAG (ChromaDB + LLaMA) + NIST / RBI / SEBI mapping

↓

4 EXECUTIVE + TECHNICAL DASHBOARD
EAL ₹18.4L • 95% VaR ₹46.2L • ROSI 32% • What-if scenarios

TECH STACK

1. Ingestion
Python, FastAPI

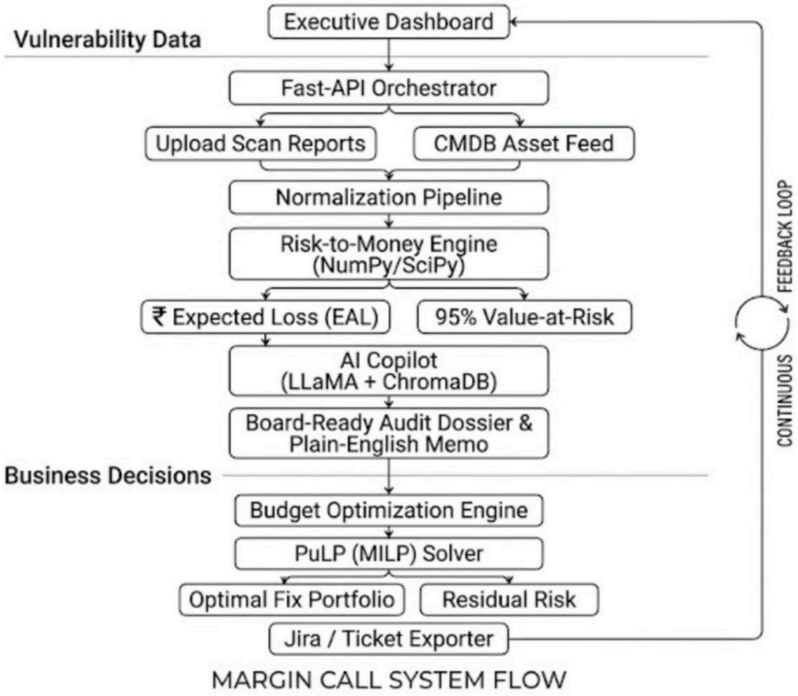
2. Risk-to-Money
NumPy, SciPy

3. Budget Optimizer
PuLP (MILP)

4. AI Copilot
LLaMA, ChromaDB

5. Executive UI
Next.js, React

WORK FLOW



MARGIN CALL SYSTEM FLOW

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FEASIBILITY AND VIABILITY

High feasibility + high market pull, without replacing existing security stacks

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TECHNICAL & OPERATIONAL

1 INDIA READY

Maps vulnerability inputs to SEBI CSCRF, RBI IT governance & cybersecurity directions, and CERT-In reporting standards.

2 GLOBAL FOUNDATION

Uses Open FAIR™ (ISO/IEC 27005), universal CVEs, FIRST EPSS and CISA KEV feeds.

3 LIGHTWEIGHT ARCHITECTURE

Stateless FastAPI engine; 10,000 Monte Carlo runs + PuLP optimization on standard cloud infrastructure; ₹0 software licensing overhead.

WHY IT IS FEASIBLE

Works with existing enterprise scanners and standard cloud infrastructure.

CHALLENGES

Complex Budget Math:
Solving combinatorial, multi-choice budget puzzles to find the absolute best fix portfolio under cap.

Audit-Safe AI:
Restricting LLM outputs to pass regulatory inspection using strict grounding, zero hallucinations, and exact clause citation.

Quantifying Loss:
Precisely modeling financial loss scenarios and statutory penalties for rare, high-impact cyber events in Rupees.

SOLUTIONS

Complex Budget Math:
PuLP MILP solver optimizes the 0/1 Knapsack problem for guaranteed optimal fix allocation under budget.

Audit-Safe AI:
LLaMA + ChromaDB RAG enforces zero hallucinations with direct RBI/SEBI clause citations.

Quantifying Loss: NumPy/SciPy Monte Carlo (10k runs) computes ₹-denominated EAL and 95% VaR.

MARKET DRIVERS & COMMERCIAL DEMAND

1 NATIONAL PULL

SEBI CSCRF, RBI Master Directions and DPDP Act compliance create demand for automated, audit-ready quantification.

2 GLOBAL MANDATES

Addresses board-accountability and operational-resilience needs under US SEC rules, EU DORA and NIS2.

3 ECONOMIC + INSURANCE IMPACT

Targets up to 40% of wasted cybersecurity spend through ROSI optimization and provides EAL / 95% VaR loss distributions.

4 MARKET DISRUPTION

Lightweight alternative to expensive legacy enterprise platforms, tailored for mid-market enterprises, fintechs and MSSPs.

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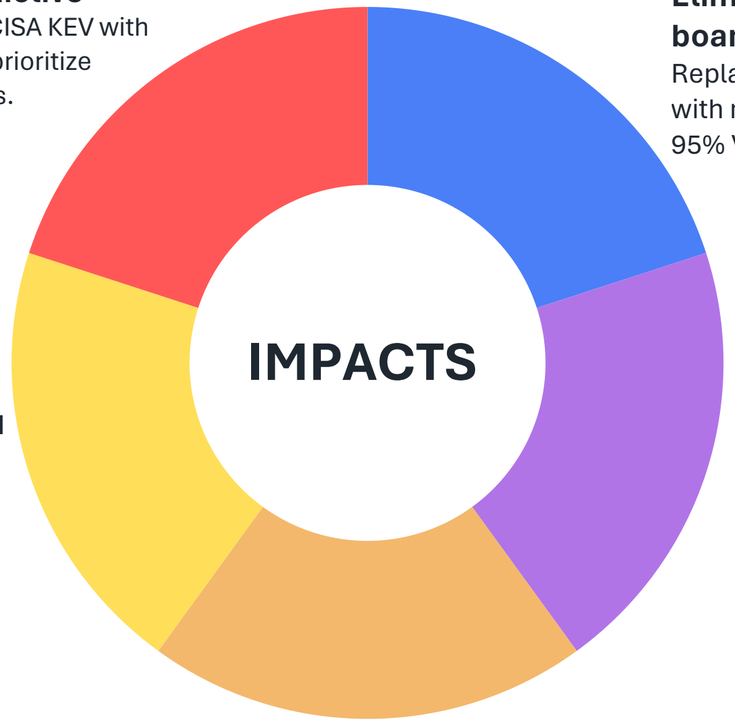
IMPACT AND BENEFITS

Moves cybersecurity from reactive alerts to measurable financial risk reduction

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Shifts cyber defense from reactive to predictive
Combines EPSS / CISA KEV with asset criticality to prioritize weaponized threats.

Eliminates the technical-to-board communication barrier
Replaces Red/Medium/High ratings with rupee loss metrics (EAL and 95% VaR).



Demonstrates measurable ROSI
Introduces efficiency frontiers and percentage yields for security tools.

Mathematically defensible budget allocation
0/1 Knapsack optimization targets vulnerabilities causing the greatest potential financial harm.

Reduces regulatory penalties and audit friction
Automates cross-referencing against RBI Master Directions, SEBI CSCRF and NIST CSF 2.0.

BENEFITS

Up to 40% reduction in wasted security spend
Prioritizes crown-jewel assets instead of low-impact fixes.

Accelerated executive decision-making
Budget slider + AI Copilot enables real-time what-if investment scenarios.

Automated noise reduction for security teams
Outputs a prioritized, mathematically ranked remediation backlog.

Improved cyber insurance posture
Generates quantitative loss distributions for premiums and policy limits.

Zero-cost, rapid deployment
Open-source architecture: Python, FastAPI, Next.js, PuLP, Groq Cloud API.

Real Life application

Client's Problem

A regional hospital network discovered 850 vulnerabilities across its Electronic Health Record (EHR) and connected diagnostic devices. Remediating every issue required ₹1.6 Cr, far surpassing their capped ₹40 Lakh cybersecurity budget while impending DPDP Act audits created massive regulatory exposure.

Metric	Before	After	Impact
Fixes	850	72	-91%
Cost	₹1.6 Cr	₹38.5L	₹1.21 Cr
Risk (EAL)	₹9.4 Cr	₹1.9 Cr	-80%
ROSI	—	19.5x	19.5x
Audit	2.5 wks	8 mins	-99%

Our Solution

Margin Call ran 10,000 Monte Carlo breach simulations to translate clinical disruption and patient privacy liability into a concrete ₹9.4 Cr Expected Annual Loss. It then applied a 0/1 Knapsack optimizer to select the 72 highest-impact remediations within the ₹40 Lakh budget, while generating audit-safe DPDP/NABH compliance dossiers via LLaMA RAG.

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RESEARCH AND REFERENCES

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RISK QUANTIFICATION

Measuring and Managing Information Risk: A FAIR Approach — Jack Freund & Jack Jones (2014)

FAIR taxonomy; Loss Event Frequency (LEF); Loss Magnitude; basis for EAL and 95% VaR.

Google Books — 2014 edition

THREAT LIKELIHOOD

Enhancing Vulnerability Prioritization: Exploit Prediction Scoring System (EPSS) — Jay Jacobs, Sasha Romanosky et al. (2023)

ML-based real-world exploitation probability; replaces static severity with dynamic exploit likelihood.

https://arxiv.org/pdf/2302.14172

BUDGET OPTIMIZATION

Selection of Optimal Countermeasure Portfolio in IT Security Planning — Tadeusz Sawik (2013)

MILP / 0-1 Knapsack formulation for selecting security countermeasures under budget limits.

https://www.sciencedirect.com/science/article/pii/S0167923613000109

GOVERNANCE & COMPLIANCE

NIST Cybersecurity Framework (CSF 2.0) — NIST SP 1300 / CSWP 29 (2024)

Cyber-risk governance, resilience and measurement; basis for compliance mapping.

https://csrc.nist.gov/pubs/sp/1300/final

LOSS CALIBRATION

Cost of a Data Breach Report — Ponemon Institute & IBM Security Research

Empirical breach benchmarks used to calibrate baseline loss distributions.

https://www.ibm.com/reports/data-breach

Margin Call vs. Existing Market Tools

Feature	Tenable / Qualys	RiskLens / Axio	Snyk / Wiz	Margin Call
How Risk is Shown	Technical scores (CVSS 1–10)	Vague high-level estimates	Code & cloud bug alerts	**Exact risk in Indian Rupees (₹)**
How Fixes are Chosen	Fix highest severity first	Manual guesswork	Fix exploitable bugs first	**Math-based optimization (Highest ROI)**
Budget Awareness	❌ Ignores budget limits	❌ No spending plan	❌ Ignores budget limits	✅ **Works strictly within your ₹ budget cap**
Audit & Compliance	Manual paperwork	Static survey templates	Basic posture checks	**Automated AI reports citing exact rules**
Action for Teams	Overwhelming list of tickets	High-level slides only	Sends raw alerts to devs	**Interactive slider + 1-click Jira tasks**